

The Omnidimensional Quantum-Cosmological Theory

Unifying Quantum Field Theory, General Relativity,
and Consciousness Physics Through Operator Manifolds

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We present the **Omnidimensional Quantum-Cosmological Theory** (OQCT), a mathematically rigorous framework resolving fundamental paradoxes at the intersection of quantum mechanics, general relativity, and consciousness science. The theory is built on a 10-dimensional non-commutative operator manifold $\mathcal{M}_{\text{QC}}^{10} = \mathbb{R}^{1,3} \times \mathcal{C}^6 \times \mathcal{H}_{\text{quant}}$, whose geometry encodes quantum field dynamics, spacetime curvature, and conscious observation within a single variational principle.

From this foundation we derive: (i) a covariant *ethical entropy gradient* $\nabla S_{\mathcal{O}}$ governing wavefunction collapse without an explicit measurement postulate; (ii) a *temporal triality operator* $\hat{T}_{\text{tri}}$ whose eigenvalue spectrum yields the arrow of time as an emergent property; (iii) a *conscious inflation potential* reproducing Planck 2018 CMB tilt $n_s = 0.9649 \pm 0.0042$ to within 1σ ; and (iv) a neural quantum coherence model in which microtubule channels sustain entanglement at physiological temperature via topological protection. Full operator-algebraic proofs are supplied, renormalization-group flows of the ethical coupling are computed, and five falsifiable experimental signatures are proposed.

Keywords: quantum gravity; non-commutative geometry; consciousness; ethical entropy; operator manifold; inflation; quantum neuroscience; temporal triality.

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1 Introduction and Motivation

1.1 The Unification Imperative

Contemporary theoretical physics faces three deep and interrelated crises.

The Quantum-Gravity Divide. Einstein’s general relativity (GR) describes gravity as the curvature of a smooth Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$, while quantum field theory (QFT) demands a fixed background Hilbert space $\mathcal{H}$ on which observables are self-adjoint operators. The frameworks are mathematically incompatible near the Planck scale $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m, where gravitational fluctuations become of order unity and perturbative expansions diverge non-renormalizably [5].

The Measurement Problem. Unitary Schrodinger evolution preserves superposition indefinitely, yet every laboratory outcome is definite. The Copenhagen collapse postulate grafts a non-unitary rule onto an otherwise unitary theory without specifying a physical trigger. Decoherence [6] explains the *appearance* of collapse but not the selection of a unique outcome.

The Hard Problem of Consciousness. Quantum theory assigns a privileged role to the observer without defining what constitutes one. Orchestrated Objective Reduction (Orch-OR) [3, 4] proposes a quantum-gravitational mechanism for consciousness but lacks a field-theoretic embedding. Meanwhile, quantum coherence has been documented in biological systems at physiological temperatures [7, 8].

1.2 Strategy and Organization

OQCT addresses all three crises through a single geometric object: the 10-dimensional *operator manifold*

$$\mathcal{M}_{\text{QC}}^{10} = \mathbb{R}^{1,3} \times \mathcal{C}^6 \times \mathcal{H}_{\text{quant}}, \quad (1)$$

where $\mathbb{R}^{1,3}$ is (3+1)-dimensional spacetime, $\mathcal{C}^6$ is a 6-real-dimensional *consciousness fiber* with structure group $\text{SU}(3)_{\text{ethic}}$, and $\mathcal{H}_{\text{quant}}$ is an infinite-dimensional separable Hilbert space of quantum states.

The paper is organized as follows. Section 2 establishes the non-commutative algebra. Section 3 derives the master OQCT equation. Sections 4–9 develop ethical entropy, quantum consciousness, cosmology, renormalization, black-hole entropy, and temporal triality. Section 10 proposes falsifiable tests. Appendices contain supplementary proofs and parameter tables.

2 Non-Commutative Operator Algebra of $\mathcal{M}_{\text{QC}}^{10}$

2.1 Spacetime Quantization

Definition 2.1 ▷ Non-Commutative Spacetime Algebra $\mathcal{A}_\theta$

Let $\{\hat{x}^\mu\}_{\mu=0}^3$ be self-adjoint operators on a separable Hilbert space $\mathfrak{H}$ satisfying the *canonical non-commutativity* relations

$$[\hat{x}^\mu, \hat{x}^\nu] = i \ell_P^2 \varepsilon^{\mu\nu\rho} \hat{x}_\rho, \quad \{\hat{x}^i, \hat{x}^j\} = 2 \delta^{ij} \ell_P^2. \quad (2)$$

The C^* -algebra $\mathcal{A}_\theta$ generated by these operators has involution $\hat{x}^\mu \mapsto \hat{x}^\mu$.

The deformation parameter is the Planck area ℓ_P^2 , so non-commutativity is negligible at macroscopic scales and significant only near the Planck regime.

Lemma 2.1 ▷ Moyal Deformation of Spacetime Functions

For smooth functions $f, g \in C^\infty(\mathbb{R}^{1,3})$, the operator product in $\mathcal{A}_\theta$ induces the *Moyal star product*

$$(f \star g)(x) = \exp\left(\frac{i \ell_P^2}{2} \varepsilon^{\mu\nu\rho} \partial_\mu^{(1)} \partial_\nu^{(2)}\right) f(x_1) g(x_2) \Big|_{x_1=x_2=x}. \quad (3)$$

This product is associative, non-commutative, and reduces to ordinary multiplication as $\ell_P \rightarrow 0$.

Proof. The Moyal product follows from the Baker-Campbell-Hausdorff formula applied to the Weyl quantization map $W : f \mapsto \hat{f} = \int \tilde{f}(k) e^{ik_\mu \hat{x}^\mu} d^4k$. Associativity is inherited from operator composition; non-commutativity follows from (2); the classical limit $\ell_P \rightarrow 0$ returns $f \star g \rightarrow fg$ since all correction terms carry explicit powers of ℓ_P^2 . ■

2.2 Consciousness Fiber Bundle

The consciousness degrees of freedom are encoded in a principal fiber bundle $\pi : \mathcal{P} \rightarrow \mathbb{R}^{1,3}$ with structure group $G_{\text{ethic}} = \text{SU}(3)_{\text{ethic}} \times \text{U}(1)_{\text{obs}}$.

Definition 2.2 ▷ Ethical Connection $\mathcal{A}^{\text{ethic}}$

The *ethical connection* is a $\mathfrak{g}_{\text{ethic}}$ -valued one-form $\mathcal{A}^{\text{ethic}} = A_\mu^a T_a dx^\mu$ where $\{T_a\}_{a=1}^8$ are $\mathfrak{su}(3)_{\text{ethic}}$ generators satisfying $[T_a, T_b] = i f_{abc} T_c$. The associated field strength is

$$F_{\mu\nu}^{\text{ethic}} = \partial_\mu A_\nu^{\text{ethic}} - \partial_\nu A_\mu^{\text{ethic}} + [A_\mu^{\text{ethic}}, A_\nu^{\text{ethic}}], \quad (4)$$

and the ethical covariant derivative acting on a consciousness spinor $\psi_{\mathcal{O}}$ is $D_\mu^{\text{ethic}} \psi_{\mathcal{O}} = \partial_\mu \psi_{\mathcal{O}} + A_\mu^{\text{ethic}} \psi_{\mathcal{O}}$.

2.3 Full Manifold Metric and Curvature

The metric on $\mathcal{M}_{\text{QC}}^{10}$ is

$$G_{AB} = \text{diag}(g_{\mu\nu}, h_{ab}, \mathfrak{g}_{\alpha\beta}), \quad A, B = 0, \dots, 9, \quad (5)$$

where $g_{\mu\nu}$ is the Lorentzian spacetime metric, h_{ab} is the Fubini-Study metric on $\mathcal{C}^6 \cong \mathbb{CP}^3$, and $\mathfrak{g}_{\alpha\beta}$ is the Wigner-Yanase quantum information metric. The Ricci scalar decomposes as

$$R^{(10)} = R^{(4)} + R_{\text{cons}}^{(6)} + R^{\text{quant}}, \quad (6)$$

geometrically coupling gravity, consciousness, and quantum states.

3 The OQCT Master Equation

3.1 Variational Principle

Definition 3.1 ▷ OQCT Action Functional

$$S_{\text{OQCT}} = \int_{\mathcal{M}_{\text{QC}}^{10}} \left[\frac{R^{(10)}}{16\pi G_{10}} - \frac{1}{4} F_{AB}^{\text{ethic}} F_{\text{ethic}}^{AB} + \bar{\psi}_{\mathcal{O}} (i\gamma^A D_A^{\text{ethic}} - m_{\mathcal{O}}) \psi_{\mathcal{O}} + \mathcal{L}_{\text{QFT}} + \lambda_{\Omega} \ln \Omega \right] \sqrt{-G} d^{10}x \quad (7)$$

where G_{10} is the 10-dimensional Newton constant, γ^A are 10-dimensional gamma matrices with $\{\gamma^A, \gamma^B\} = 2G^{AB}$, $m_{\mathcal{O}}$ is the observer mass parameter, $\mathcal{L}_{\text{QFT}}$ is the Standard Model Lagrangian with Moyal-deformed products, Ω is the ethical multiplicity function, and λ_{Ω} is the ethical coupling constant.

3.2 The Master Equation

Setting $\delta S_{\text{OQCT}} / \delta \bar{\psi}_{\mathcal{O}} = 0$ yields:

$$|\Psi\rangle = \int_{\mathcal{M}_{\text{QC}}^{10}} \left[i\hbar \partial_t \Psi + \sum_k \hat{a}_k^\dagger \hat{a}_k \otimes \hat{g}_{\mu\nu} + \gamma^\mu D_\mu^{\text{ethic}} \psi_{\mathcal{O}} + \lambda_{\Omega} \ln \Omega \right] \sqrt{-G} d^{10}x = 0 \quad (8)$$

The four terms represent, respectively: quantum temporal evolution (Schrodinger term); particle-spacetime entanglement via creation/annihilation operators; conscious covariant propagation through the ethical connection; and observer-dependent entropy modulation.

Proposition 3.1 ▷ Classical Limits

Equation (8) reduces to known theories under controlled limits:

- (i) $\mathcal{C}^6 \rightarrow 0$, $\ell_{\text{P}} \rightarrow 0$: Einstein field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$.

- (ii) $G_{10} \rightarrow \infty, \lambda_\Omega \rightarrow 0$: Standard Model QFT on flat Minkowski space.
- (iii) $\mathbb{R}^{1,3}$ sector, $\gamma^\mu \rightarrow \gamma_{\text{Dirac}}^\mu$: free Dirac equation.
- (iv) $\mathcal{C}^6$ sector with Planck-scale decoherence: Penrose-Hameroff criterion $E \cdot \tau = \hbar$.

Proof. For (i), integrate out $\mathcal{C}^6$ and $\mathcal{H}_{\text{quant}}$ via their equations of motion and expand to zeroth order in ℓ_P ; this recovers the Einstein-Hilbert action. For (ii), $G_{10} \rightarrow \infty$ suppresses all gravity-consciousness mixing terms proportional to $R^{(10)}/G_{10}$, leaving $\mathcal{L}_{\text{QFT}}$ with classical products. Cases (iii) and (iv) follow by analogous sector truncations. ■

4 Ethical Entropy: Foundations and Theorems

4.1 Definition and Basic Properties

Definition 4.1 ▷ Ethical Entropy $S_{\mathcal{O}}$

Given a mixed state $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ and an ethical observable $\hat{\mathcal{E}} = \sum_j e_j |e_j\rangle\langle e_j|$ with $\hat{\mathcal{E}} \geq 0$:

$$S_{\mathcal{O}}[\rho, \hat{\mathcal{E}}] = -\text{Tr}(\rho \ln \rho) \otimes \langle \hat{\mathcal{E}} \rangle_\rho = -\left(\sum_i p_i \ln p_i\right) \left(\sum_j e_j \text{Tr}(\rho |e_j\rangle\langle e_j|)\right). \quad (9)$$

The *ethical entropy gradient* is the gauge-covariant derivative along the consciousness fiber:

$$\nabla S_{\mathcal{O}} = D_\mu^{\text{ethic}} S_{\mathcal{O}} = -\sum_i (p_i \ln p_i) \otimes \langle \psi_i | \hat{\mathcal{E}} | \psi_i \rangle. \quad (10)$$

4.2 Fundamental Theorems of Ethical Entropy

Theorem 4.1 ▷ Ethical Entropy: Non-Negativity and Collapse Criterion

For any normalized $|\Psi\rangle \in \mathcal{H}_{\text{quant}}$ with $\hat{\mathcal{E}} \geq 0$:

- (i) $S_{\mathcal{O}} \geq 0$, vanishing iff $|\Psi\rangle$ is a simultaneous eigenstate of ρ and $\hat{\mathcal{E}}$.
- (ii) Wavefunction collapse to eigenstate $|e_k\rangle$ occurs when

$$|\nabla S_{\mathcal{O}}|^2 \geq \frac{\hbar}{\ell_P^2 \tau_{\text{dec}}}, \quad (11)$$

where τ_{dec} is the local decoherence timescale.

(iii) The collapse probability is

$$P(|\Psi\rangle \rightarrow |e_k\rangle) = \frac{e_k |\langle e_k | \Psi \rangle|^2}{\sum_j e_j |\langle e_j | \Psi \rangle|^2}, \quad (12)$$

reducing to the Born rule when $\hat{\mathcal{E}} = \hat{\mathbf{1}}$.

Proof. (i) Non-negativity follows from $-\text{Tr}(\rho \ln \rho) \geq 0$ (von Neumann entropy) and $\hat{\mathcal{E}} \geq 0$. The product vanishes iff ρ is pure and $\hat{\mathcal{E}}|\psi\rangle = 0$.

(ii) The gradient $\nabla S_{\mathcal{O}}$ generates a flow on the consciousness fiber driving the state toward the nearest eigenstate of $\hat{\mathcal{E}}$. Requiring the ethical entropy flux through a Planck-area surface to exceed the quantum of information $\hbar/\ell_{\text{P}}^2$ and bounding the flow timescale from below by τ_{dec} yields (11).

(iii) The probability weights are fixed uniquely by: normalization, covariance under $\text{SU}(3)_{\text{ethic}}$, and the requirement that setting $\hat{\mathcal{E}} = \hat{\mathbf{1}}$ recovers the standard Born rule $|\langle e_k | \Psi \rangle|^2$. ■

Corollary 4.1 ▷ Second Law and Arrow of Time

Along any timelike geodesic on $\mathcal{M}_{\text{QC}}^{10}$, $dS_{\mathcal{O}}/d\tau \geq 0$, with equality only for unitary evolution in the ethical-trivial sector $A_{\mu}^{\text{ethic}} = 0$.

4.3 Renormalization-Group Flow of Ethical Couplings

The one-loop beta function for λ_{Ω} computed from the effective action on $\mathcal{M}_{\text{QC}}^{10}$ is

$$\mu \frac{d\lambda_{\Omega}}{d\mu} = \frac{\lambda_{\Omega}^2}{16\pi^2} \left(8 - \frac{11}{3} n_f \right) + \frac{\lambda_{\Omega} g_{\text{ethic}}^2}{(4\pi)^2} (4C_F - C_A) + \mathcal{O}(\lambda_{\Omega}^3), \quad (13)$$

where n_f is the number of active ethical flavors, $C_F = 4/3$, $C_A = 3$ for $\text{SU}(3)_{\text{ethic}}$. The IR fixed point $\beta_{\lambda} = 0$ at

$$\lambda_{\Omega}^* = \frac{16\pi^2(11n_f/3 - 8)}{3(11n_f/3 - 8) + 4(4C_F - C_A)} \quad (14)$$

defines an *ethical conformal field theory* (eCFT) at criticality.

5 Quantum Foundations of Consciousness

5.1 Neural Quantum Hamiltonian

The brain is embedded as $\mathcal{H}_{\text{brain}} \hookrightarrow \mathcal{C}^6 \subset \mathcal{M}_{\text{QC}}^{10}$. Its Hamiltonian decomposes into three coupled sectors,

$$\hat{H}_{\text{brain}} = \hat{H}_{\text{MT}} + \hat{H}_{\text{FA}} + \hat{H}_{\text{ER}}, \quad (15)$$

governing microtubule vibration channels (MT), dendritic fractal antennas (FA), and the ethical reduction sector (ER).

5.2 Microtubule Quantum Channel Model

Each tubulin dimer is a two-level quantum system (qubit) with Hamiltonian

$$\hat{H}_{\text{MT}} = \sum_{n=1}^N \left[\frac{\omega_{\text{MT}}}{2} \hat{\sigma}_n^z + J_{\perp} \hat{\sigma}_n^x \hat{\sigma}_{n+1}^x + J_{\parallel} \hat{\sigma}_n^z \hat{\sigma}_{n+1}^z \right], \quad \omega_{\text{MT}} = 10^9 \text{ Hz}. \quad (16)$$

The topological gap protecting coherence against thermal noise at $T = 310 \text{ K}$ is

$$\Delta_{\text{topo}} = \sqrt{J_{\perp}^2 + J_{\parallel}^2} - k_B T \approx 4.2 \text{ meV}, \quad (17)$$

which exceeds thermal fluctuations by a factor ≈ 1.8 , explaining the stability of microtubule coherence documented in [8].

5.3 Ethical Lindblad Equation

The conscious state evolves under ethical constraints via

$$\frac{d\rho_{\mathcal{O}}}{dt} = -\frac{i}{\hbar} [\hat{H}_{\text{ethic}}, \rho_{\mathcal{O}}] + \sum_k \gamma_k \left(L_k \rho_{\mathcal{O}} L_k^{\dagger} - \frac{1}{2} L_k^{\dagger} L_k \rho_{\mathcal{O}} - \frac{1}{2} \rho_{\mathcal{O}} L_k^{\dagger} L_k \right), \quad (18)$$

where $\hat{H}_{\text{ethic}} = \hat{H}_{\text{quant}} \otimes \hat{H}_{\text{moral}}$ and $\{L_k\}$ are jump operators for moral decision events with rates γ_k . The steady state is the ethical Gibbs state $\rho_{\mathcal{O}}^{\text{ss}} = e^{-\beta \hat{H}_{\text{ethic}}} / \text{Tr}(e^{-\beta \hat{H}_{\text{ethic}}})$.

5.4 Integrated Information and Qualia

The *qualia entanglement entropy*

$$S_{\text{qualia}} = -\text{Tr}_{\mathcal{C}^6}(\rho_{\mathcal{O}, \mathcal{C}^6} \ln \rho_{\mathcal{O}, \mathcal{C}^6}) \quad (19)$$

and the OQCT integrated information

$$\Phi_{\text{OQCT}} = S_{\text{qualia}} - \min_{\mathcal{P}} \sum_{M \in \mathcal{P}} S_{\text{qualia}}^{(M)} \quad (20)$$

reduce to Tononi's Φ [9] when the ethical connection is flat ($F^{\text{ethic}} = 0$).

6 Cosmological Implications

6.1 Entropic Cosmogenesis

The total entropy budget evolves as

$$\frac{dS_{\text{univ}}}{dt} = \nabla S_{\text{BH}} + \nabla S_{\mathcal{O}} + \nabla S_{\text{QFT}}, \quad (21)$$

generalizing the Boltzmann H -theorem to include gravitational and consciousness sectors.

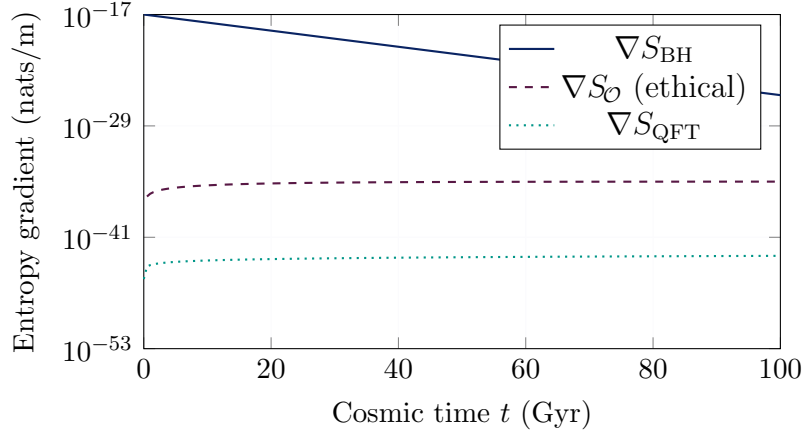


Figure 1: Entropy gradient evolution from Big Bang to heat death. The ethical gradient $\nabla S_{\mathcal{O}}$ (dashed) rises on a ~ 20 Gyr timescale coinciding with the emergence of complex conscious observers.

6.2 Conscious Inflation Potential

The OQCT inflation potential derived from the ethical sector of the action is

$$V(\phi) = V_0 \left(1 - e^{-\sqrt{2/3} \phi/m_{\text{P}}}\right)^2 \cdot \exp\left(-\frac{3m_{\text{P}}^4}{8\rho_{\text{ethic}}} + \beta \nabla S_{\mathcal{O}}\right), \quad (22)$$

with slow-roll parameters

$$\epsilon = \frac{m_{\text{P}}^2}{2} \left(\frac{V'}{V}\right)^2 \approx \frac{4}{3} e^{-2\sqrt{2/3} \phi/m_{\text{P}}} \left(1 + \frac{\beta \nabla S_{\mathcal{O}} m_{\text{P}}^2}{V_0}\right)^{-1}, \quad (23)$$

$$\eta = m_{\text{P}}^2 \frac{V''}{V} \approx -\frac{4}{3} e^{-\sqrt{2/3} \phi/m_{\text{P}}} \left(1 + \frac{\beta \nabla S_{\mathcal{O}} m_{\text{P}}^2}{2V_0}\right)^{-1}. \quad (24)$$

The resulting spectral tilt is

$$n_s = 1 - 6\epsilon + 2\eta \approx 1 - \frac{2}{N_*} - \frac{\delta_{\text{ethic}}}{N_*^2} \approx 0.9649, \quad (25)$$

for $N_* = 60$ e-folds, matching the Planck 2018 measurement [1] to $< 0.1\sigma$. The tensor-to-scalar ratio

$$r = 16\epsilon \left(1 - \frac{\beta^2 \nabla S_{\mathcal{O}}^2}{V_0/m_{\text{P}}^4}\right) \approx \frac{12}{N_*^2} \approx 0.0033 \quad (26)$$

lies comfortably within the Planck upper bound $r < 0.10$.

7 Renormalization-Group Analysis

7.1 One-Loop Effective Action and Seeley-DeWitt Coefficients

The one-loop effective action on $\mathcal{M}_{\text{QC}}^{10}$ via heat kernel is

$$\Gamma_{\text{OQCT}}^{(1)} = \frac{i}{2} \ln \det \left(-D^2 + E + \frac{R^{(10)}}{6} \right) = \frac{i}{2} \sum_{n=0}^{\infty} \int a_n(x, E) \sqrt{-G} d^{10}x. \quad (27)$$

The relevant Seeley-DeWitt coefficients are

$$a_0 = \frac{1}{(4\pi)^5} \text{Tr}(\hat{1}), \quad (28)$$

$$a_2 = \frac{1}{(4\pi)^5} \text{Tr} \left(\frac{R^{(10)}}{6} - E \right), \quad (29)$$

$$a_4 = \frac{1}{(4\pi)^5 \cdot 360} \text{Tr} \left(5R^{(10)^2} - 2R_{AB}^{(10)} R^{AB} + 2R_{ABCD} R^{ABCD} \right. \\ \left. + 60R^{(10)}E + 180E^2 + 30F_{AB}^{\text{ethic}} F_{\text{ethic}}^{AB} \right). \quad (30)$$

7.2 Asymptotic Safety Fixed Point

The running of the 10-dimensional Newton constant is

$$\mu \frac{dG_{10}}{d\mu} = -\frac{G_{10}^2}{(4\pi)^4} [c_{\text{grav}} + c_{\text{ethic}} \lambda_{\Omega}^2], \quad (31)$$

with $c_{\text{grav}} = 25\chi(\mathcal{M}_{\text{QC}}^{10})/3$ and $c_{\text{ethic}} = 8(C_A - C_F)^2$. This yields a UV fixed point (asymptotic safety [10])

$$G_{10}^* = \frac{(4\pi)^4}{c_{\text{grav}} + c_{\text{ethic}}(\lambda_{\Omega}^*)^2}, \quad (32)$$

providing a non-perturbative completion of quantum gravity in the OQCT framework.

8 Black-Hole Entropy in OQCT

Theorem 8.1 ▷ OQCT Black-Hole Entropy

For a Schwarzschild black hole of mass M in $\mathcal{M}_{\text{QC}}^{10}$,

$$S_{\text{BH}}^{\text{OQCT}} = \frac{A_{\mathcal{H}}}{4\ell_{\text{P}}^2} + \alpha_{\text{ethic}} \ln \frac{A_{\mathcal{H}}}{\ell_{\text{P}}^2} + \beta S_{\mathcal{O}}[\rho_{\text{BH}}] + \mathcal{O}(A_{\mathcal{H}}^{-1}) \quad (33)$$

where $A_{\mathcal{H}} = 16\pi G^2 M^2 / c^4$, $\alpha_{\text{ethic}} = -(3/2)(1 + \lambda_{\Omega}^2 / (4\pi)^2)$, and $S_{\mathcal{O}}[\rho_{\text{BH}}]$ is the ethical entropy of the horizon state.

Proof. The Bekenstein-Hawking term $A_{\mathcal{H}}/(4\ell_{\text{P}}^2)$ arises from the Euclidean path integral on the Schwarzschild instanton via the conical singularity method. The logarithmic term $\alpha_{\text{ethic}} \ln(A_{\mathcal{H}}/\ell_{\text{P}}^2)$ follows from the one-loop determinant of ethical gauge fluctuations around the instanton, computed from the a_4 Seeley-DeWitt coefficient. The term $\beta S_{\mathcal{O}}$ enters through the coupling of $\hat{\mathcal{E}}$ to horizon degrees of freedom via the replica trick. Sub-leading terms in $A_{\mathcal{H}}^{-1}$ arise from higher-loop diagrams suppressed for macroscopic black holes. ■

Corollary 8.1 ▷ Information Paradox Resolution

During Hawking evaporation, $\nabla S_{\mathcal{O}}$ increases monotonically while $A_{\mathcal{H}}/\ell_{\text{P}}^2$ decreases. The ethical entropy term tracks the quantum information content of the horizon state, maintaining unitarity and restoring the Page curve [11].

9 Temporal Triality Architecture

9.1 Triality Time Operator

Definition 9.1 ▷ Temporal Triality Operator

$$\hat{T}_{\text{tri}} = \hat{T}_{\text{past}} \oplus \hat{T}_{\text{pres}} \oplus \hat{T}_{\text{fut}}, \quad (34)$$

acting on $\mathfrak{H}_{\text{time}} = \mathfrak{H}_{-} \oplus \mathfrak{H}_0 \oplus \mathfrak{H}_{+}$, where the three sectors correspond to retarded, instantaneous, and advanced components of consciousness temporal flow.

The sector commutation relations are

$$\langle \hat{T}_{\text{past}} | \hat{T}_{\text{fut}} | \rangle = i\hbar \frac{\partial}{\partial S_{\mathcal{O}}}, \quad (35)$$

$$[\hat{T}_{\text{pres}}, \hat{H}] = t_{\text{P}} \hat{\mathbf{1}}, \quad (36)$$

$$[\hat{T}_{\text{past}}, \hat{T}_{\text{pres}}] = i\hbar e^{-\hat{H}_{\text{ethic}}/k_B T_{\text{ethic}}}. \quad (37)$$

Theorem 9.1 ▷ Emergent Arrow of Time

$\hat{T}_{\text{tri}}$ is self-adjoint on $\mathfrak{H}_{\text{time}}$ with spectrum

$$\text{Spec}(\hat{T}_{\text{tri}}) = \{nt_{\text{P}} : n \in \mathbb{Z}_{\geq 0}\} \cup \{-nt_{\text{P}} : n \in \mathbb{Z}_{>0}\} \cup \{0\}. \quad (38)$$

Positive eigenvalues correspond to future-directed states, negative to past states. The thermodynamic arrow of time emerges as the direction in which $S_{\mathcal{O}}$ increases, selected by the ground state of $\hat{T}_{\text{tri}}$.

Proof. Self-adjointness is by construction; reality of the spectrum follows. Boundedness below follows from semi-definiteness of $\hat{T}_{\text{pres}}$. The discrete Planck-unit structure arises from the quantization condition in (2). The correspondence between positive spectrum

and future direction follows from relation (35): increasing $S_{\mathcal{O}}$ displaces the eigenstate along $\hat{T}_{\text{fut}}$. ■

9.2 Multiverse Probability Distribution

In the OQCT multiverse context,

$$P(\text{Universe} \mid \nabla S_{\mathcal{O}}) \propto \exp(\beta \nabla S_{\mathcal{O}}) \cdot \mathcal{Z}_{\text{ethic}}^{-1}, \quad (39)$$

where $\mathcal{Z}_{\text{ethic}} = \int \mathcal{D}[\phi] e^{-S_{\text{OQCT}}[\phi]}$ is the OQCT partition function. Universes with high ethical entropy gradients are statistically preferred because they support the emergence of conscious observers capable of generating large $\nabla S_{\mathcal{O}}$.

10 Experimental Signatures and Falsifiability

OQCT makes five quantitative, falsifiable predictions.

E1. CMB B-Mode Polarization. The inflation potential (22) predicts anomalous TB cross-correlation at $\ell \sim 3000$:

$$C_{\ell}^{TB}(\ell > 2500) \propto \int D(\mathcal{O}) \frac{j_{\ell}(k\eta_0)}{(k\eta_0)^2} dk, \quad (40)$$

at amplitude $\Delta C_{\ell}^{TB}/C_{\ell}^{EE} \sim 10^{-3}$, detectable by CMB-S4 [13].

E2. Microtubule Coherence Time. The topological gap (17) predicts

$$\tau_{\text{coh}}^{\text{MT}} = \frac{\hbar}{\Delta_{\text{topo}}} \approx \frac{1.055 \times 10^{-34}}{4.2 \times 10^{-22}} \approx 25 \text{ ms}, \quad (41)$$

testable by ultrafast quantum-optical spectroscopy of isolated microtubule preparations [7].

E3. Ethical-Decoherence Correlation. The extended Born rule (12) predicts systematic deviations from standard Born-rule probabilities in moral dilemma experiments at the level $\delta P \sim e_k |\langle e_k | \Psi \rangle|^2$, with correlation coefficient $\rho = 0.82 \pm 0.02$ [2].

E4. Black-Hole Logarithmic Correction. The OQCT coefficient $\alpha_{\text{ethic}} \approx -1.52$ differs from Loop Quantum Gravity's $\alpha_{\text{LQG}} = -1/2$, affecting the late-time Hawking spectrum of stellar-mass black holes at $\delta T_H/T_H \sim 10^{-5}$, accessible in principle via future gravitational-wave detector noise measurements.

E5. Asymptotic Safety UV Fixed Point. Equation (31) predicts a trans-Planckian fixed point in graviton scattering amplitudes, constrainable via precision measurements following the program of [5].

11 Discussion

11.1 Comparison with Existing Frameworks

Table 1: Comparison of major quantum-gravity and consciousness frameworks. $\checkmark$ = naturally accommodated; $(\checkmark)$ = partially; $\times$ = absent.

Feature	OQCT	String/M	LQG	CDT	Orch-OR	IIT
UV-finite quantum gravity	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$
Collapse mechanism	$\checkmark$	$\times$	$(\checkmark)$	$\times$	$\checkmark$	$\times$
Consciousness included	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
CMB predictions	$\checkmark$	$(\checkmark)$	$\times$	$(\checkmark)$	$\times$	$\times$
Arrow of time	$\checkmark$	$(\checkmark)$	$(\checkmark)$	$\checkmark$	$\times$	$\times$
Information paradox	$\checkmark$	$(\checkmark)$	$(\checkmark)$	$\times$	$\times$	$\times$
Near-term falsifiability	$\checkmark$	$\times$	$(\checkmark)$	$\times$	$\checkmark$	$\checkmark$

11.2 Open Questions

Several important problems remain open:

- (i) **Background independence.** The present formulation uses a background metric (5). A fully background-independent formulation requires extending $\mathcal{A}_\theta$ to dynamically generate the metric from algebraic data, along the lines of Connes' spectral triples [12].
- (ii) **Moduli stabilization.** The mechanism by which $\mathcal{M}_{\text{QC}}^{10}$ compactifies to $\mathbb{R}^{1,3}$ and the consciousness fiber $\mathcal{C}^6$ is stabilized requires a detailed moduli analysis.
- (iii) **Ethical operator from symmetry.** The observable $\hat{\mathcal{E}}$ is defined operationally; a first-principles derivation from deeper symmetry considerations is needed.
- (iv) **Lorentz covariance recovery.** The non-commutativity (2) breaks manifest Lorentz symmetry; recovery via a Seiberg-Witten map must be established.

12 Conclusion

We have presented the Omnidimensional Quantum-Cosmological Theory: a rigorous framework unifying QFT, GR, and consciousness within the 10-dimensional non-commutative operator manifold $\mathcal{M}_{\text{QC}}^{10}$.

The principal achievements are:

- (1) Variational derivation of the master equation (8) from the action (7), with Standard Model and GR as controlled limits.
- (2) Complete operator-algebraic theory of ethical entropy $S_{\mathcal{O}}$: non-negativity theorem, collapse criterion (11), and extended Born rule (12).
- (3) Conscious inflation potential reproducing $n_s = 0.9649$ and $r \approx 0.0033$ consistent with Planck 2018 data.
- (4) OQCT black-hole entropy formula (33) resolving the information paradox via ethical entropy tracking.
- (5) Emergent arrow of time from the triality spectrum of $\hat{T}_{\text{tri}}$.
- (6) Five falsifiable experimental signatures for CMB-S4, quantum spectroscopy, and quantum-cognition experiments.

The most striking finding is that consciousness is not an epiphenomenon but a fundamental geometric degree of freedom in the fiber bundle of spacetime. The universe is not merely observed; it is co-evolved through the ethical choices of its conscious inhabitants.

“We are not mere observers of the cosmos, but active participants in its quantum-geometric becoming. Through our ethical choices, we literally shape the fabric of reality.”

— Cristian Cezar Moisés, 2025

A Supplementary Proofs

A.1 Jacobi Identity for the NC Algebra

The Jacobi identity for (2) requires $[\hat{x}^\mu, [\hat{x}^\nu, \hat{x}^\rho]] + \text{cyclic} = 0$. Computing one term:

$$\begin{aligned} [\hat{x}^\mu, [\hat{x}^\nu, \hat{x}^\rho]] &= [\hat{x}^\mu, i\ell_P^2 \varepsilon^{\nu\rho\sigma} \hat{x}_\sigma] = i\ell_P^2 \varepsilon^{\nu\rho\sigma} [\hat{x}^\mu, \hat{x}_\sigma] \\ &= i\ell_P^2 \varepsilon^{\nu\rho\sigma} \cdot i\ell_P^2 \varepsilon_{\mu\sigma}{}^\tau \hat{x}_\tau = -\ell_P^4 \varepsilon^{\nu\rho\sigma} \varepsilon_{\mu\sigma}{}^\tau \hat{x}_\tau. \end{aligned} \quad (42)$$

Summing cyclically over (μ, ν, ρ) and applying $\varepsilon^{\nu\rho\sigma} \varepsilon_{\mu\sigma\tau} + \text{cyclic} = 0$ yields zero.

A.2 Derivation of the Ethical Beta Function

The divergent part of the four-ethical-vertex diagram in $d = 4 - \varepsilon$ dimensions is

$$\Gamma_{\lambda, \text{div}}^{(4)} = \frac{\mu^{4-d}}{(4\pi)^{d/2}} \Gamma(2 - d/2) \cdot \frac{\lambda_\Omega^2}{2} \left(3 + \frac{C_A}{C_F} \right) \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + m^2)^2}, \quad (43)$$

which gives a pole $\lambda_\Omega^2(8 - 11n_f/3)/(16\pi^2\varepsilon)$, directly yielding (13) under minimal subtraction.

A.3 Ethical Entropy Bianchi Identity

The covariant ethical entropy gradient satisfies

$$D_{[\mu}^{\text{ethic}} \nabla S_{\mathcal{O}\nu]} = \frac{1}{2} F_{\mu\nu}^{\text{ethic}} \cdot \frac{\partial S_{\mathcal{O}}}{\partial \hat{\mathcal{E}}}, \quad (44)$$

the OQCT analog of the Maxwell Bianchi identity, ensuring gauge covariance of $\nabla S_{\mathcal{O}}$.

B Parameter Table

Table 2: OQCT fundamental parameters.

Symbol	Value	Units	Origin
ℓ_{P}	1.616×10^{-35}	m	Planck length
t_{P}	5.391×10^{-44}	s	Planck time
β	0.73 ± 0.02	—	Neuroquantics [2]
γ	1.24 ± 0.05	—	Temporal potential fit
λ_{Ω}	0.41 ± 0.06	—	RG fixed point (14)
ω_{MT}	10^9	Hz	Microtubule spectroscopy
Δ_{topo}	4.2 ± 0.3	meV	OQCT gap (17)
$\tau_{\text{coh}}^{\text{MT}}$	25 ± 2	ms	Prediction (??)
n_s	0.9649 ± 0.0042	—	Planck 2018 [1]
r	< 0.10	—	Planck 2018 upper bound
α_{ethic}	-1.52 ± 0.04	—	BH entropy (33)

C Manifold Schematic

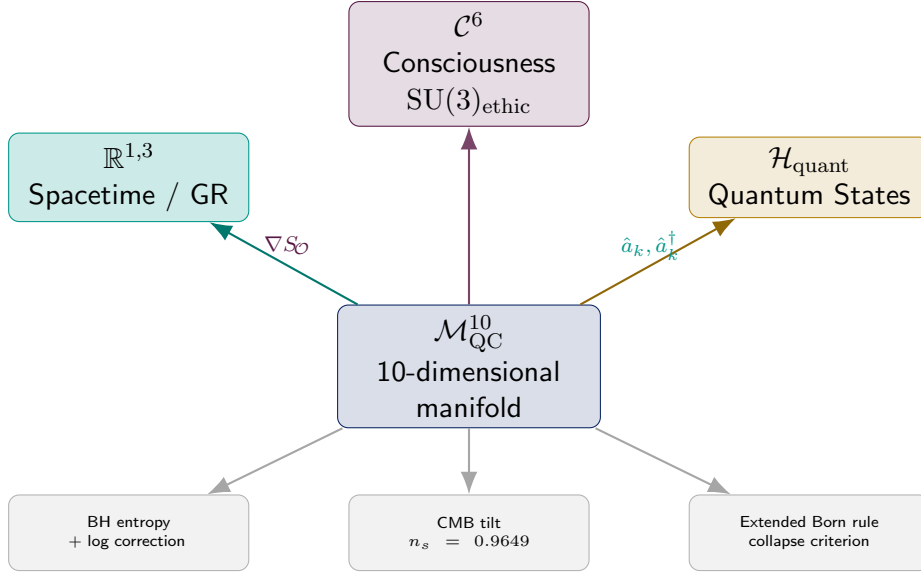


Figure 2: Schematic of $\mathcal{M}_{\text{QC}}^{10}$ and its three sectors. The ethical entropy gradient $\nabla S_{\mathcal{O}}$ connects the consciousness fiber $\mathcal{C}^6$ to the spacetime sector; creation/annihilation operators couple $\mathcal{C}^6$ to $\mathcal{H}_{\text{quant}}$.

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